THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The brightness of a bulb increases if: 1

(a) more cells are added in series **(b)** a cell is removed
(c) the switch is opened **(d)** the wire is cut

2. Which of these is a reason for using aluminium in power lines? 1

(a) heavy **(b)** insulator
(c) brittle **(d)** light and a good conductor

3. The materials that make the best connecting wires are: 1

(a) wood and rubber **(b)** plastic and glass
(c) copper and aluminium **(d)** paper and cloth

4. What happens if one of two bulbs connected in series burns out? 1

(a) the cell is drained **(b)** the other goes off too
(c) the other glows more **(d)** the other glows normally

5. Which of the following is the symbol of a closed switch? 1

(a) a zigzag **(b)** a circle
(c) a gap **(d)** a line joining two points

6. In a circuit tester, the bulb glows when the object between the two ends is: 1

(a) copper wire **(b)** plastic **(c)** wood **(d)** rubber

7. In which device is the heating effect not wanted? 1

(a) geyser **(b)** laptop **(c)** iron **(d)** toaster

8. A fuse wire should have ___ melting point than the copper wire. 1

(a) a lower **(b)** no **(c)** the same **(d)** a higher

9. Which of the following conducts electricity? 1

(a) salt solution **(b)** pure water **(c)** sugar solution **(d)** kerosene

10. Why are filament bulbs becoming less popular? 1

(a) too cheap **(b)** too bright
(c) waste much energy as heat **(d)** too cold

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How is an LED different from a filament bulb in the way it glows? 2

12. Why do birds sitting on a bare power line not get a shock? 2

13. Why does a thin wire get hotter than a thick wire with the same current? 2

14. Why should electrical appliances be switched off from the socket when not in use? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a bulb in a torch does not glow when the batteries are put in the wrong way. 3

16. How can you make a bulb brighter without changing the bulb? 3

17. Which two things should be checked if a bulb does not glow in your circuit? 3

OR

Why does a fuse blow when an electric heater and an iron are used together on the same socket? 3

18. Why should damaged plugs and wires be replaced immediately? 3

OR

Differentiate between fuse and switch. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how a fuse and an MCB protect a home and why a wrong fuse is dangerous. 5

OR

Describe the factors that affect heating in a wire when current passes. 5

20. Describe the construction of a torch, its circuit and why it stops working. 5

OR

Describe the path of electricity inside a house from the main switch to an appliance. 5

END OF QUESTION PAPER — Total: 40 Marks

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